

المحاضرة ٩

طرق إيجاد التكامل

$$\int_{y=g(x)}^{\boxed{F(x)}} \boxed{dx}$$

التكامل بالتعويض !

$$\int \frac{\sqrt{x+2}}{x+6} dx$$

أوجد التكاملات التالية !

الحل

$$\text{let } t^2 = x+2$$

$$2t dt = dx$$

$$\int \frac{\sqrt{x+2}}{x+6} dx = \int \frac{\sqrt{t^2}}{t^2+4} \cdot 2t dt$$

$$= \int \frac{2t^2}{t^2+4} dt$$

$$= 2 \int \frac{t^2}{t^2+4} dt$$

$$= 2 \int \frac{t^2+4-4}{t^2+4} dt = 2 \int 1 - \frac{4}{t^2+4}$$

$$= 2t - 8 \int \frac{1}{t^2+4}$$

$$= 2t - 8 \left[\frac{1}{2} \tan^{-1} \frac{t}{2} \right] + C$$

$$= 2t - 4 \tan^{-1} \left(\frac{t}{2} \right) + C$$

$$= 2\sqrt{x+2} - 4 \tan^{-1} \frac{\sqrt{x+2}}{2} + C$$

الاجابة

$$\int (x^2+1)^2 \cdot x \cdot dx$$

$$\frac{1}{2} \int 2(x^2+1)^2 \cdot x$$

$$= \frac{(x^2+1)^3}{3} + C$$

$$\int \frac{1}{x^2+a^2} dx \quad \text{--- / 2135$$

$$= \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

$$\frac{1}{\sqrt{a^2-x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right)$$

$$\frac{1}{x\sqrt{x^2-a^2}} dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{x^2}{(x-1)^4} dx$$

$$\text{let } (x-1) = u$$

$$dx = du$$

$$\int \frac{x^2}{(x-1)^4} dx = \int \frac{(1+u)^2}{u^4} du$$

$$\int \frac{(1+u^2+2u)}{u^4} du$$

$$\int \left[\frac{1}{u^4} + \frac{u^2}{u^4} + \frac{2u}{u^4} \right] du$$

$$\int u^{-4} + u^{-2} + 2u^{-3} du$$

$$= \frac{u^{-3}}{-3} + \frac{u^{-1}}{-1} + \frac{2u^{-2}}{-2} du$$

$$= \frac{u^{-3}}{-3} - u^{-1} - u^{-2} du$$

$$= \frac{(x-1)^{-3}}{-3} - (x-1)^{-1} - (x-1)^{-2} dx$$

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$$\int \frac{dx}{\sqrt{x} - \sqrt[3]{x}}$$

$$\text{let } t = \sqrt{x}$$

$$dt = \frac{1}{2\sqrt{x}} dx$$

$$2t dt = dx$$

$$\int \frac{dx}{\sqrt{x} - \sqrt[3]{x}} = \int \frac{2t dt}{t - (t)^3}$$

$$\int \frac{2 dt}{1 - (t)^2} = 2 \int \frac{dt}{1 - (t)^2}$$

$$= 2 \tanh^{-1}(t) + C \quad \#$$

$$= 2 \tanh^{-1} \sqrt{x} + C$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{a} \tanh^{-1} \frac{x}{a} + C \quad \left(\frac{1}{a} \right)$$

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \sinh^{-1} \frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \frac{x}{a} + C$$

التحويلات الثلاثية

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$$\sqrt{a^2 - x^2}$$

$$x = a \sin u$$

$$u = \sin^{-1} \frac{x}{a}$$

$$\sqrt{a^2 + x^2}$$

$$x = a \tan u$$

$$u = \tan^{-1} \frac{x}{a}$$

$$\sqrt{x^2 - a^2}$$

$$x = a \sec u$$

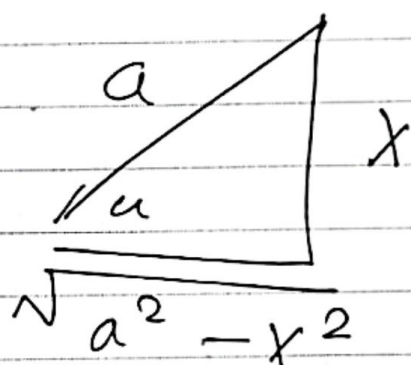
$$u = \sec^{-1} \frac{x}{a}$$

$$\sin^2 u + \cos^2 u = 1$$

$$1 - \sin^2 u = \cos^2 u$$

$$1 + \tan^2 u = \sec^2 u$$

$$\sec^2 u - 1 = \tan^2 u$$



$$\sin u = \frac{x}{a}$$

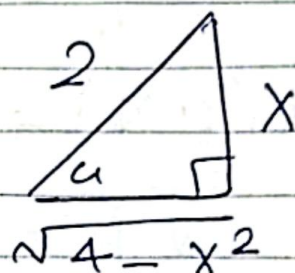
$$\cos u = \frac{\sqrt{a^2 - x^2}}{a}$$

$$\tan u = \frac{x}{\sqrt{a^2 - x^2}}$$

$$\int \frac{x^2}{\sqrt{4-x^2}} dx$$

$$x = 2 \sin u$$

$$dx = 2 \cos u \cdot du$$



$$= \int \frac{4 \sin^2 u}{\sqrt{4-2\sin^2 u}} \cdot 2 \cos u du$$

$$= \int \frac{4 \sin^2 u}{\cancel{2} \sqrt{1-\sin^2 u}} \cdot \cancel{2} \cos u du$$

$$= \int \frac{4 \sin^2 u}{\cancel{\cos u}} \cdot \cancel{\cos u} du$$

$$= \int 4 \sin^2 u du$$

$$= 2 \int [1 - \cos 2u] du$$

$$2 \left[u - \frac{1}{2} \sin 2u \right] + C$$

$$2u - \sin 2u \cos u + C \quad \neq$$

تذكر
 $\cos 2x = 1 - 2\sin^2 x$
 $2\sin^2 x = 1 - \cos 2x$

التكامل بالتجزئ

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$$d(uv) = u dv + v du$$

$$\int u dv = \int d(uv) - \int v du$$

$$\boxed{\int u dv = uv - \int v du} \quad \#$$

Ex: $\int (\ln x) dx$

$$u = \ln x \quad du = \frac{1}{x} dx$$

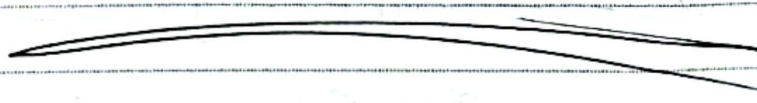
$$du = \frac{1}{x} dx \quad v = x$$

$$\int \ln x dx = \int u dv$$

$$= uv - \int v du$$

$$= x \ln x - \int x \cdot \frac{1}{x} dx$$

$$= x \ln x - x + C \quad \#$$



$$\int \tan^{-1}(x) dx$$

$$u = \tan^{-1}(x)$$

$$dv = dx$$

$$du = \frac{1}{1+x^2} dx$$

$$v = x$$

$$\int \tan^{-1}(x) = \int u dv$$

$$= uv - \int v du$$

$$= x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx$$

$$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$$

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